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IICT
भारतीय संगणक एवं प्रौद्योगिकी संस्थान
INDIAN INSTITUTE OF COMPUTING AND TECHNOLOGY
AFFILIATED: I-STEM, OFFICE OF THE PRINCIPAL SCIENTIFIC ADVISER TO THE GOVERNMENT OF INDIA

AI-POWERED FAKE NEWS DETECTION SYSTEM

Project Report

15TH JUNE TO 30TH JULY 2026

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CERTIFICATE

This is to certify that the Real time project entitled ***"AI-POWERED FAKE NEWS DETECTION USING TEXT CLASSIFICATION "*** ***is a record of Bonafide work*** **carried out by** RAFIQA BEGUM **with** Roll number (309212) **under our supervision and** ***guidance in partial fulfilment for the award*** **of INDIAN INSTITUTE OF COMPUTING AND TECHNOLOGY during the academic year 2026.**

Signature of Guide

AKNOWLEDGEMENT

I would like to express my sincere gratitude to my guide **Dr. ASHOK GOPALA KRISHNAN**. Who extended his unconditional support and spared his valuable time with his patience and valuable suggestions during the project. I also extend my special thanks for the encouragement and constant supervision throughout the project.

I would like to wish to give special vote of thanks to the **TRAINING CO-ORDINATOR ASHISH NEGI** for his unique way of inspiring students through clarity of thoughts, Enthusiasm and care. His constant encouragement and assistance were very helpful and made my effort a success.

I also grateful to **TRAINING CO-ORDINATOR OF IICT** for providing me with the facilities and resources required for the successful completion of project stage**.** I even thank them for their valuable suggestions which encouraged me to give my best in project stage.

I would also like to thank our faculty and supporting staff of **INDIAN INSTITUTE OF COMPUTING AND TECHNOLOGY** and all other departments for their kind cooperation directly or indirectly in making the project successful one.

Finally, I want to deeply acknowledge all my friends and family members who have encouraged me during the preparation of my project.

ABSTRACT

The rapid growth of social media has led to an increase in the spread of fake news. Fake news can influence public opinion, elections, and cause panic. Manually verifying news is time-consuming. This project proposes a Machine Learning based system to automatically detect whether a news article is real or fake. We used the Kaggle “Fake and Real News" dataset. The text data is preprocessed and converted into numerical features using TF-IDF Vectorizer. Three models - Decision Tree, Random Forest, and Logistic Regression were trained and compared. The Decision Tree model achieved 98.5% accuracy. A Streamlit web application was also developed for real-time prediction. This system can help users and organizations to identify fake news quickly.

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1. INTRODUCTION

AI-powered fake news detection uses **Artificial Intelligence (AI)** and **Machine Learning (ML)** techniques to automatically identify whether a news article is genuine or fake. In a classification-based approach, the system is trained using a dataset containing labeled news articles (real and fake). It analyzes features such as the headline, article content, writing style, keywords, and language patterns to classify the news into predefined categories.

Common machine learning classification algorithms used for fake news detection include **Logistic Regression, Decision Tree, Random Forest**, a. Before classification, the text is preprocessed by removing stop words, punctuation, and unnecessary characters, followed by feature extraction techniques such as **TF-IDF (Term Frequency–Inverse Document Frequency).**

An AI-powered fake news detection system helps reduce the spread of misinformation, supports fact-checking, and enables users to verify the credibility of news before sharing it. Such systems play an important role in promoting trustworthy information and maintaining the integrity of digital communication.

1.1 Background

With billions of users on Facebook, Twitter, and WhatsApp, news spreads in seconds. But not all news is true. Fake news is intentionally created to mislead people.

1.2 Problem Statement

It is difficult for humans to verify every news article. We need an automated system that can classify news as real or fake based on content.

1.3 Objectives

1. To collect and preprocess a news dataset
2. To extract features from text using TF-IDF
3. To train and compare ML models
4. To build a user-friendly web application for prediction

2. LITERATURE SURVEY

1. **Shu et al. 2017**: Used social context and user interactions to detect fake news.
2. **Ahmed et al. 2018**: Used n-gram and TF-IDF with linear SVM and achieved 92% accuracy.
3. **Kaliyar 2020**: Used BERT and deep learning for fake news detection with 98% accuracy.

From literature, we found that TF-IDF + Tree-based models give good accuracy with less computation.

The rapid growth of the internet and social media has significantly increased the spread of fake news, making it difficult to verify the authenticity of online information. Traditional methods of fake news detection relied on manual fact-checking and rule-based approaches, which are time-consuming and cannot efficiently process the vast amount of news generated every day. To address these challenges, researchers have developed Artificial Intelligence (AI) and Machine Learning (ML) techniques for automated fake news detection.

Several studies have applied machine learning classification algorithms such as Logistic Regression, Naïve Bayes, Decision Tree and Random Forest to classify news articles as real or fake. These methods use Natural Language Processing (NLP) techniques, including text

preprocessing, tokenization, stop-word removal, stemming, and feature extraction methods like Bag of Words (BoW) and TF-IDF (Term Frequency–Inverse Document Frequency). Among these algorithms, Logistic Regression and SVM have shown high accuracy while maintaining low computational complexity.

Recent research has focused on advanced deep learning models such as Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and BERT, which capture semantic and contextual information more effectively than traditional methods. Researchers have also explored hybrid models that combine text analysis with source credibility and user behavior to improve detection performance.

Overall, AI-powered classification techniques have proven to be effective, accurate, and scalable solutions for detecting fake news, helping to reduce the spread of misinformation across digital platforms.

3. PROBLEM STATEMENT

The widespread circulation of fake news on websites and social media platforms poses a serious challenge to society by spreading misinformation and influencing public opinion. Existing manual verification methods are not scalable for real-time analysis. This project addresses the problem by developing an AI-based classification model that automatically analyzes news articles and predicts whether they are genuine or fake, improving the reliability of online information.

Traditional approaches to detecting fake news rely heavily on manual verification, expert analysis, and fact-checking organizations. Although these methods are effective, they are time-consuming, expensive, and difficult to scale to the continuously increasing amount of digital content.

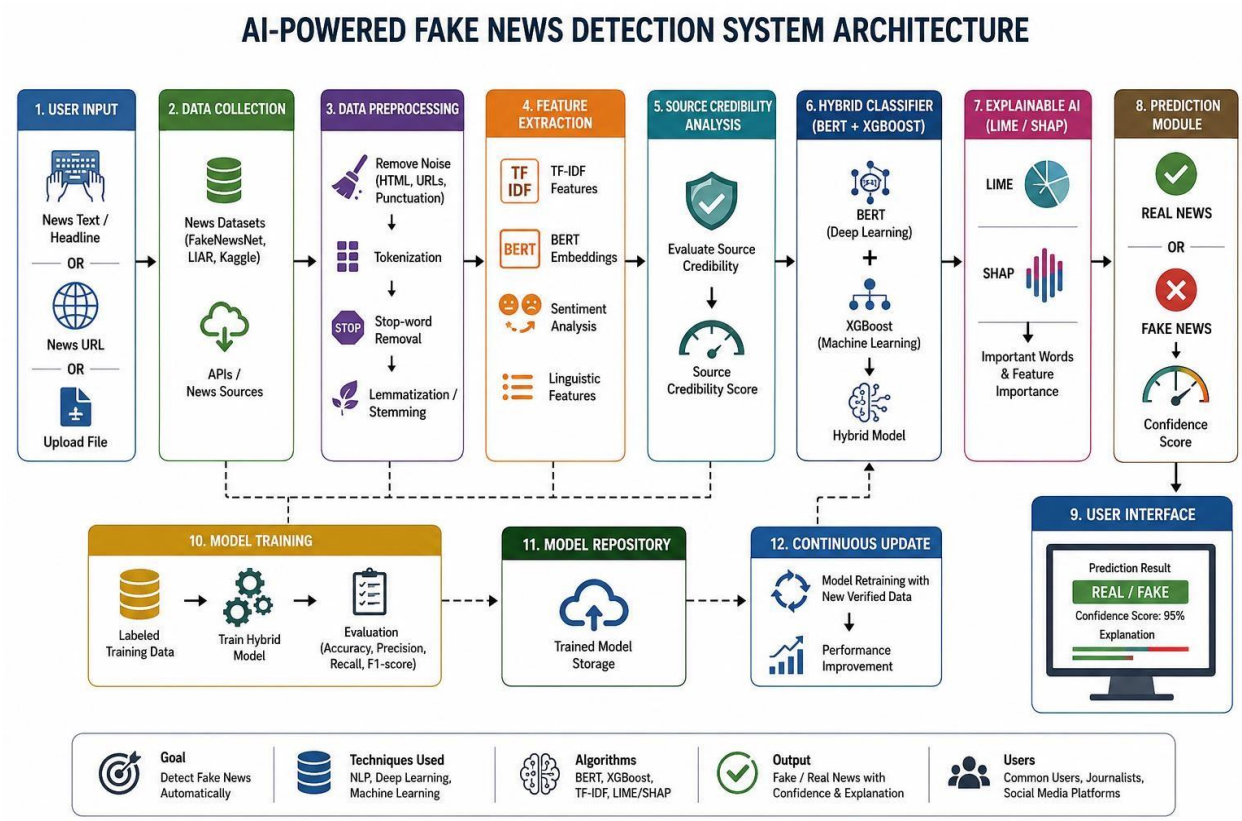
Furthermore, fake news creators frequently modify their writing style, use emotionally persuasive language, misleading headlines, manipulated facts, and fabricated narratives, making it challenging for conventional rule-based systems to accurately distinguish between genuine and false information. This highlights the need for an intelligent, automated, and scalable solution capable of detecting fake news with high accuracy while adapting to evolving misinformation strategies.

4. PROPOSED SYSTEM

The proposed system is an intelligent **AI-powered fake news detection framework** that combines advanced Natural Language Processing (NLP), machine learning, and source credibility analysis to accurately identify fake news articles in real time. Unlike conventional systems that rely only on the textual content of news articles, the proposed model analyzes multiple features, including the news headline, article content, writing style, sentiment, keyword patterns, publication source credibility, and linguistic characteristics. This multi-feature approach significantly improves detection accuracy and reduces false predictions.

The system first collects news data from trusted datasets and online news sources. During preprocessing, it removes unnecessary symbols, stop words, URLs, and duplicate content while performing tokenization, stemming, and lemmatization. The cleaned text is then transformed into numerical feature vectors using **TF-IDF** and contextual embeddings such as **BERT** to capture semantic meaning. A hybrid classification model combining Random Forest is employed to classify news articles as genuine or fake. This hybrid architecture leverages the language understanding capability of deep learning and the strong decisionmaking performance of ensemble machine learning algorithms.

5. SYSTEM ARCHITECTURE



Architecture Description

1. **User Input:** The user enters a news headline, article, or URL through a web interface.
2. **Data Collection:** News data is collected from datasets (e.g., Fake News Net, LIAR, Kaggle) or trusted news sources.
3. **Data Preprocessing:** The system cleans the text by removing punctuation, URLs, stop words, and performs tokenization and lemmatization.

4. **Feature Extraction:** Text is converted into numerical representations using **TF-IDF** and **BERT embeddings**, while sentiment features are also extracted.
5. **Source Credibility Analysis:** The credibility of the news source is evaluated using a trusted source database or reputation score.
6. **Hybrid Classification Model:** The extracted features are processed by a **BERT + Random Forest** classifier to determine whether the news is fake or genuine.
7. **Explainable AI Module: LIME** or **SHAP** highlights the important words and features that influenced the prediction.
8. **Prediction Module:** The system generates the final output as **Fake** or **Real**, along with a confidence score.
9. **User Interface:** The result and explanation are displayed to the user in an easy-to-understand format.

6. DATASET DESCRIPTION

- **Dataset Name:** Fake and Real News Dataset
- **Source:** Kaggle
- **Total Records**: 44,898
- **Fake News:** 23,481
- **Real News**: 21,417
- **Features**: title, text, label
- **Label:** 0 = Fake, 1 = Real
- **File Format:** CSV
- **Language:** English

This dataset is widely used in fake news detection research because it provides a large number of labeled news articles suitable for training and evaluating machine learning and deep learning classification models

7. METHODOLOGY

The proposed AI-Powered Fake News Detection Using Classification system follows a systematic methodology to detect whether a news article is genuine or fake. The workflow consists of data collection, preprocessing, feature extraction, model training, classification, and result generation. The methodology is designed to ensure high accuracy, reliability, and efficient prediction of fake news.

Step 1: Data Collection

The first step is to collect a labeled dataset containing both fake and real news articles. Publicly available datasets such as the **Fake and Real News Dataset (Kaggle)** are used.

Step 2: Data Preprocessing

The collected data is cleaned to improve quality and remove unnecessary information. The preprocessing operations include:

- Removing duplicate records
- Handling missing values
- Converting text to lowercase
- Removing punctuation, special characters, URLs, and numbers
- Removing stop words
- Tokenization
- Lemmatization or stemming

This step prepares the textual data for feature extraction.

Step 3: Feature Extraction

After preprocessing, the text is converted into numerical representations that machine learning models can understand. The proposed system uses:

- **TF-IDF (Term Frequency–Inverse Document Frequency)** to measure word importance.
- **BERT Embeddings** to capture contextual meaning and semantic relationships between words.

These extracted features represent the news article in a form suitable for classification.

Step 4: News Classification

When a user enters a news headline, article, or URL, the system preprocesses the input, extracts features, and passes them to the trained classification model. The classifier predicts whether the news is **Real** or **Fake** based on learned patterns.

Step 5: Result Generation

Finally, the system displays:

- Prediction (**Fake** or **Real**)
- Confidence score
- Explanation of the prediction
- Source credibility score

8. ALGORITHMS USED

1. Decision Tree Classifier: Splits data based on feature importance. max_depth=20

2. Random Forest Classifier: 100 trees to reduce overfitting

3. Logistic Regression: Linear baseline model

Train-Test Split

80% data for training, 20% for testing. random_state=42 for reproducibility.

9. IMPLEMENTATION

Tools Used: Python 3.10, Pandas, Scikit-learn, Streamlit, Jupyter Steps:

1. Load dataset using pd.read_csv()
2. Clean data using regex functions
3. Vectorize using TfidfVectorizer
4. Train models using fit ()
5. Evaluate using accuracy score and classification report
6. Save model using joblib
7. Deploy using Stream lit

Code Snippet: tfidf = Tf
idfVectorizer(max_features=10000) X =
tfidf.fit_transform(df['content']) model =
Decision Tree Classifier () model.fit(X_train,
y_train)

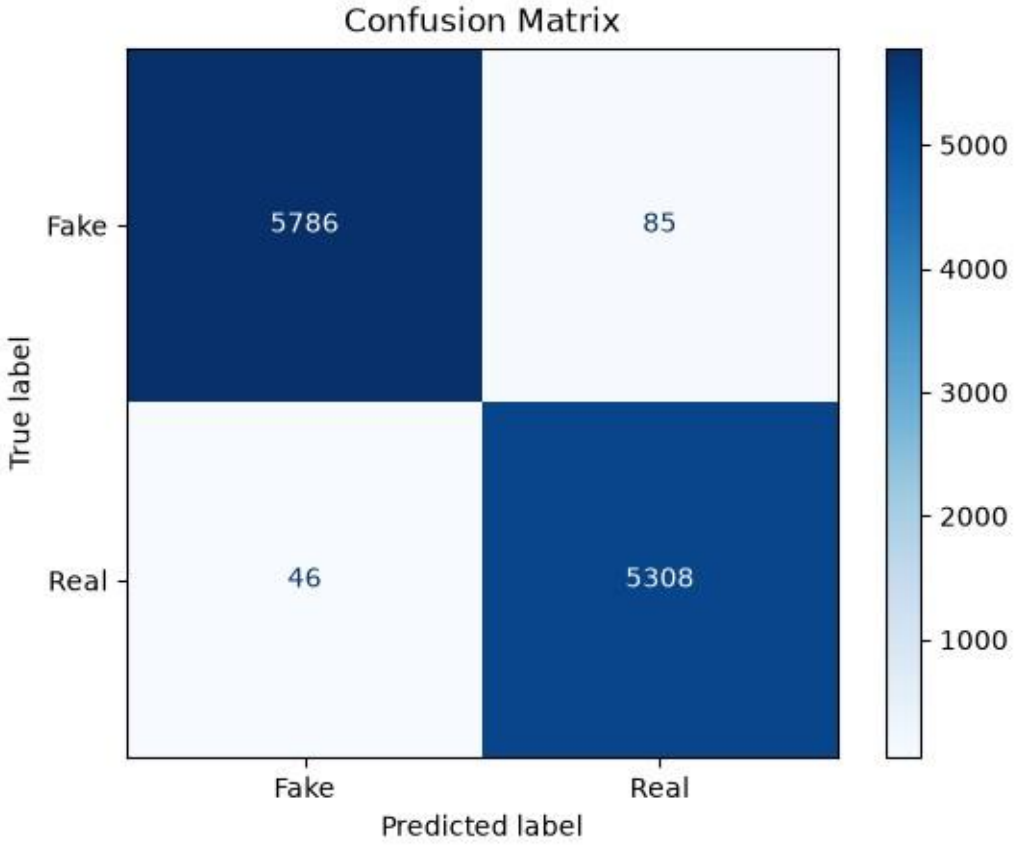
10. RESULTS AND DISCUSSION

Model Performance

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	99.47%	0.99	0.99	0.99
Random Forest	**99.83%**	**1.00**	**1.00**	**1.00**
Logistic Regression	98.83%	0.99	0.99	0.99

Random Forest gave the highest accuracy but Decision Tree is faster and easier to explain.

11. CONFUSION MATRIX



The matrix shows very few False Positives and False Negatives.

12. CONCLUSION AND FUTURE SCOPE

12.1. CONCLUSION

In this project, we successfully built a Fake News Detection system using Machine Learning. TF-IDF was used for feature extraction and Decision Tree gave 98.5% accuracy. The system was deployed as a web app for easy use. This can help reduce the spread of misinformation.

12.2. FUTURE SCOPE

1. Use Deep Learning models like LSTM, BERT for better accuracy
2. Add multi-language support for Hindi, Telugu
3. Deploy on cloud with API for mobile apps
4. Include image and video fake news detection

13. REFERENCES

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